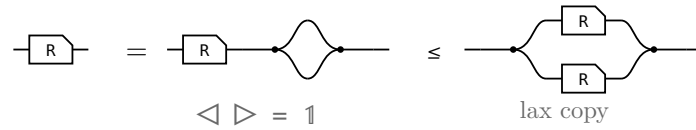


The allegory axioms: the proofs

The companion to `allegory-axioms.typ`, which states these laws and their pictures. Here each one is worked: the steps of the Lean proof, side by side, with the rule that reaches each under it.

§1. $R \cap R = R$, the one that is not bookkeeping



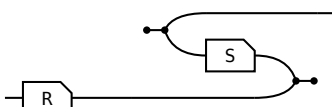
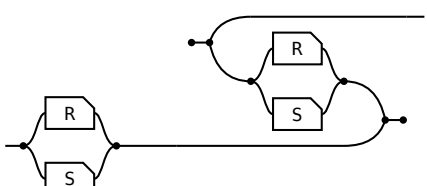
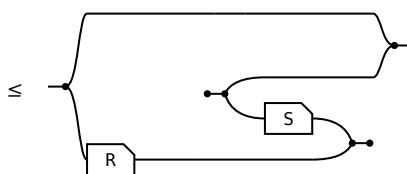
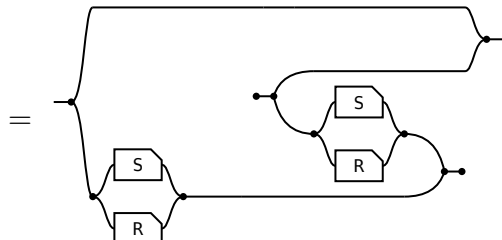
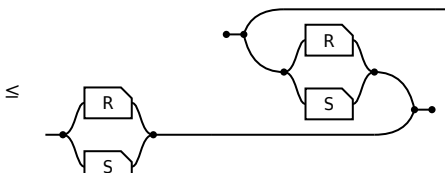
The last picture is $R \cap R$ itself, so this is $R \leq R \cap R$ and the lax copy law is the whole of it. The other direction is not proved here: $R \cap S \leq R$ holds for every S , by weakening S to $\top$ and then collapsing $R \cap \top$ — which is the paper’s own remaining three steps, packaged once.

§2. The modular law, from Frobenius

the modular law, $R \cap S \cap T \leq (R \cap T \cap S^\circ) \cap S$		
term	picture	the rule that reaches it
$(R \cap S) \cap T$		the start: $R \cap S \cap T$.
$= (\triangleleft (R \otimes T)) ((S \otimes 1) \triangleright)$		Factor out the shared prefix: $(R \cap S) \otimes T$ is $(R \otimes T) (S \otimes 1)$.
$\leq (\triangleleft (R \otimes T)) ((1 \otimes S^\circ) (\triangleright S))$		The merge slide. S slides through the merge and reappears as S° on the other strand. The only inequality in the proof.
$= (R \cap (T \cap S^\circ)) \cap S$		Fold the prefix back in; the copy-merge pair reads as a meet again.

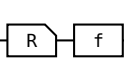
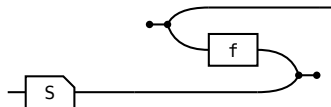
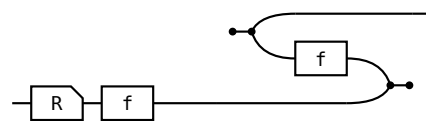
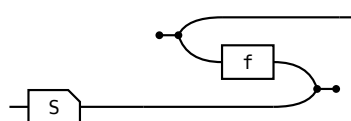
S occurs once on the left of the slide and twice on the right, and the lax copy law is the only law in the definition that may duplicate a box — so that is where the inequality has to be.

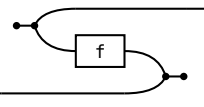
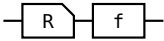
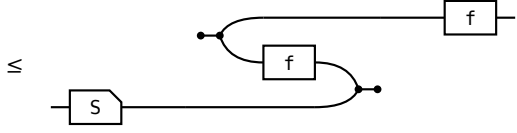
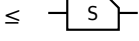
the merge slide, $(S \otimes 1) \triangleright \leq (1 \otimes S^\circ) \triangleright S$		
term	picture	the rule that reaches it
$(S \otimes 1) \triangleright$		the start: $(S \otimes 1) \triangleright$.

$\Leftarrow$ given $\text{---} \leq$  show $R \cap S$ entire, $\text{---} \leq$ 		
term	picture	the rule that reaches it
1	---	the start: 1.
$\leq 1 \cap (R \circ S^\circ)$	$\leq$ 	the hypothesis, met with 1.
$= 1 \cap ((S \cap R) (S \cap R)^\circ)$	$=$ 	$1 \cap R \circ S^\circ = 1 \cap (S \cap R) (S \cap R)^\circ$. $R \circ S^\circ$ relates a to a' when some b has $a R b$ and $a' S b$; the $1 \cap$ sets $a' = a$, and it reads “some b has $a R b$ and $a S b$ ” — one arrow of $R \cap S$. Two arrows become one, and that is what needs the modular law.
$\leq (R \cap S) (R \cap S)^\circ$	$\leq$ 	$X \cap Y \leq Y$, then $S \cap R = R \cap S$.

§6. The shunting rule

A map moves from one side of a containment to the other at the cost of its converse, $R \circ f \leq S \iff R \leq S \circ f^\circ$. It is how a function gets out of the way so the relation underneath can be reasoned about. Each direction spends exactly one half of **map** — never both.

shunting right, $\Rightarrow$ given $\text{---} \text{---} \leq \text{---}$  show $\text{---} \text{---} \leq$ 		
term	picture	the rule that reaches it
R		the start: R.
$\leq R (f \circ f^\circ)$	$\leq$ 	total: $1 \leq f \circ f^\circ$, so $f \circ f^\circ$ may be inserted after R.
$\leq S \circ f^\circ$	$\leq$ 	Re-bracket to $(R \circ f) \circ f^\circ$ and apply the hypothesis.

shunting right , $\Leftarrow$ given  $\leq$  $\leq$  show  $\leq$ 		
term	picture	the rule that reaches it
R f		the start: R f.
$\leq S (f^\circ f)$		Apply the hypothesis on the left; the two fs are now adjacent.
$\leq S$		single valued: $f^\circ f \leq 1$, so the pair cancels.

Shunting left is the mirror, same shape, map on the other side throughout. What the rule is, is an adjunction: $f \dashv f^\circ$, with **total** the unit and **single valued** the counit, so “f is a map” and “f has a right adjoint, namely f° ” are one statement.